Omid Tarabavar

Email: omidtarab1300@gmail.com
Mobile: +98-9939338694
GitHub: github.com/omidTarabavar

Research Interests

- **Multimodal Machine Learning**: Exploring multimodal representation learning and fusion techniques to integrate textual, visual, and contextual information, particularly for fake news detection and fact verification.
- **Social Computing**: Studying misinformation and information credibility in online communities, with an interest in how social context and user interactions influence the spread and detection of misleading content.
- **Large Language Models and Agentic AI**: Evaluation and benchmarking of LLMs, with an interest in the design and optimization of agentic systems, including planning, memory, and task routing.
- **Trustworthy Machine Learning**: Investigating model reliability and robustness under distribution shifts, with a particular focus on domain generalization to unseen data and environments.

Education

Amirkabir University of Technology Tehran, Iran

M.Sc. in Artificial Intelligence & Robotics, **GPA: 18.86/20 | Current Class Rank: 1st** Oct 2025 - Present

Master's Thesis: Fake News Detection Using Multimodal RAG and Learned Query Representation

Advisors: Dr. Momtazi and Dr. Safabakhsh

- Description: Developing a multimodal RAG framework that learns query representations for evidence retrieval and combines textual and visual information to assess news credibility.

Selected Coursework: Natural Language Processing (20/20), Machine Learning (20/20), Deep Learning (18.2/20), Trustworthy Machine Learning (18.33/20), Advanced Natural Language Processing (17.7/20).

Shiraz University of Technology Shiraz, Iran

B.Sc. in Computer Engineering and Information Technology Sep 2021 - Sep 2025

GPA: 19.09/20 (4.00/4.00) | Graduated 1st in Class [Transcript]

Bachelor's Thesis: Typing Assistant [Report]

- Description: Developed a next-word prediction system using probabilistic models (n-grams), sequence models (LSTMs), and attention-based models (Transformers) to enhance text completion accuracy.

Selected Coursework: Artificial Intelligence (20/20), Fundamentals of Computational Intelligence (20/20), Linear Algebra (20/20), Engineering Probability and Statistics (19/20), Algorithm Design (19.25/20).

Publications

Visual-Anchored Context Fusion for Source-Faithful Evidence Retrieval in Multimodal Fact Verification.
Omid Tarabavar, Saeedeh Momtazi, and Reza Safabakhsh.
Status: Submitted to *Applied Soft Computing* (awaiting review).
Summary: Proposes VACF, a training-free multimodal evidence retrieval framework that combines protected visual-anchor selection, contextual completion, and source-faithful restoration to preserve evidence provenance.

Selected M.Sc. Course Projects

- **Persian Banking Assistant: Hybrid Intent Routing, RAG, and LLM-as-a-Judge** [Code] Advanced NLP, 2026
  - Built a multi-turn Persian banking chatbot in which a fine-tuned ParsBERT intent classifier routes low-confidence queries to a few-shot LLM fallback (accuracy 87.7% $\rightarrow$ 96.2%), with LLM-based slot filling and hybrid dense + BM25 retrieval with cross-encoder reranking.
  - Evaluated the system with RAGAS and a custom Gemma-based LLM-as-a-judge, showing that the two agree on answer correctness ($r = 0.74$) but barely on faithfulness ($r = 0.19$); tested improvements with paired significance tests.
- **Parameter-Efficient Fine-Tuning and Catastrophic Forgetting in Gemma 3n** [Code] Advanced NLP, 2026
  - Fine-tuned Gemma 3n on Persian news with LoRA and with bottleneck adapters implemented from scratch in PyTorch. The adapters used 64% fewer trainable parameters, reached lower in-domain perplexity (5.97 vs. 6.06), and forgot less out of domain (+19.9% vs. +33.6% Wikipedia perplexity).
- **Adversarial Robustness and Fairness Audit** [Code] Trustworthy ML, 2026
  - Implemented FGSM/PGD attacks and adversarial training on Fashion-MNIST and CIFAR-10, and ruled out gradient masking with strong PGD, EOT-PGD, transfer attacks, and AutoAttack (CIFAR-10 AutoAttack accuracy 0% $\rightarrow$ 30.7%).
  - Audited income classifiers on Adult for group and intersectional fairness, calibration, SHAP explanations, and counterfactual recourse, and compared the trade-offs of reweighing and threshold-based mitigation.
- **Generalization Under Distribution Shift** [Code] Trustworthy ML, 2026
  - Compared importance reweighting, DANN, and CycleGAN for unsupervised domain adaptation on rotated MNIST, showing that all fail under severe (90°) shift, and studied layer-wise transfer from 200 labeled target samples (80.2% from scratch $\rightarrow$ 96.1%).

Teaching Experience

- **Teaching Assistant** at Sharif University of Technology
  - Natural Language Processing (Fall 2026, *Graduate course*) — Dr. Mahdi Samiei
- **Teaching Assistant** at Amirkabir University of Technology
  - Deep Learning (Fall 2026, *Graduate course*) — Dr. Reza Safabakhsh
- **Teaching Assistant** at Shiraz University of Technology
  - Algorithm Design (Spring 2024, *Undergraduate course*) — Dr. Omid Bushehrian
  - Formal Languages and Automata (Spring 2024, *Undergraduate course*) — Dr. Ali Sadeghi
  - Data Structures (Fall 2023, *Undergraduate course*) — Dr. Rasool Esmaeilyfard
  - Advanced Programming (Spring 2023, *Undergraduate course*) — Dr. Pirooz Shamsinejad
- **Instructor for ICPC Preparation** at Shiraz University of Technology
  - Covered fundamentals of competitive programming to help students prepare for the 25th Asian Regional ICPC Contest in Iran (Fall 2024).

Working Experience

- **Modern Cybernetics** — Shiraz, Iran
  On-site, Machine Learning Developer — Jan 2025 - Jun 2025
  - Description: Developed AI agents using fine-tuned large language models (LLMs) and Retrieval-Augmented Generation (RAG), integrating data retrieved from web sources through automated scraping tools.
- **Council of Artificial Intelligence Scientific Association** — Shiraz, Iran
  Shiraz University of Technology, Deputy Secretary — May 2023 - Jul 2024
  - Description: Organized and facilitated AI-related events, workshops, and competitions for students, including preparations for competitive programming contests such as ICPC.

Honors and Awards

- **Admitted to the M.Sc. program in Artificial Intelligence** at Amirkabir University of Technology through direct admission based on academic excellence.
- **Ranked within the top 2%** among 126,000+ participants in the National University Entrance Exam.
- **Admitted to NODET High Schools** (National Organization for Development of Exceptional Talents) through the competitive entry exam.

Online Courses

- **Linear Algebra (18.06)** [Notes] – *MIT, Gilbert Strang* — Jul 2025 - Sep 2025
- **Deep Learning Specialization** –*DeepLearning.AI, Andrew Ng et al.* — Feb 2025 - Mar 2025
- **Natural Language Processing Specialization** –*DeepLearning.AI, Łukasz Kaiser et al.* — Sep 2024 - Oct 2024
- **Machine Learning Specialization** [Certificate] –*DeepLearning.AI, Andrew Ng et al.* — Sep 2024 - Oct 2024
- **Mathematics for Machine Learning and Data Science Specialization** – *DeepLearning.AI* — Jul 2024 - Aug 2024

Skills Summary

- **Programming Languages**: Python, Java, C++, C
- **Libraries and Frameworks**: TensorFlow, Keras, PyTorch, Scikit-Learn, Pandas, NumPy, NLTK, Matplotlib
- **Tools**: Git, MySQL, LaTeX
- **Systems and Environments**: Windows, Linux, Jupyter Notebook
- **Soft Skills**: Time Management, Teaching, Adaptability, Collaboration, Leadership, Critical Thinking
- **English**: IELTS mock test (Aug 2024): Overall 7.0 (unofficial)

References

- **Dr. Saeedeh Momtazi** – Associate Professor, Amirkabir University of Technology — momtazi@aut.ac.ir
- **Dr. Reza Safabakhsh** – Professor, Amirkabir University of Technology — safa@aut.ac.ir
- **Dr. Mahdi Samiei** – Sharif University of Technology — mm.samiei@sharif.edu
- **Dr. Omid Bushehrian** – Associate Professor, Shiraz University of Technology — bushehrian@sutech.ac.ir
- **Dr. Ali Sadeghi** – Assistant Professor, Shiraz University of Technology — sadeghi@sutech.ac.ir
- **Dr. Rasool Esmaeilyfard** – Assistant Professor, Shiraz University of Technology — esmaeily@sutech.ac.ir
- **Dr. Pirooz Shamsinejad** – Assistant Professor, Shiraz University of Technology — p.shamsinejad@sutech.ac.ir